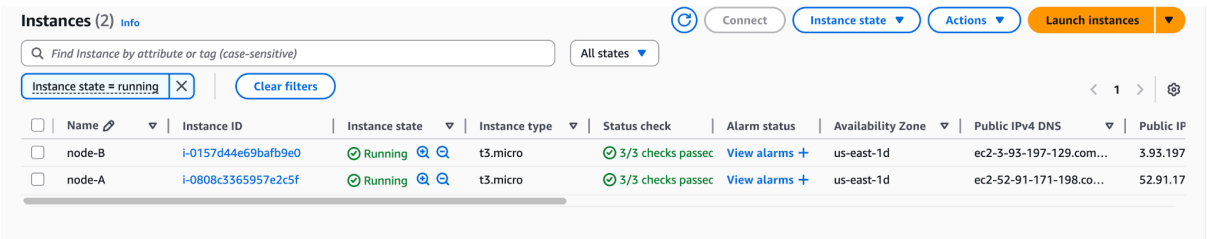
**DC lab1 – Danial Baitakov IT-2307**

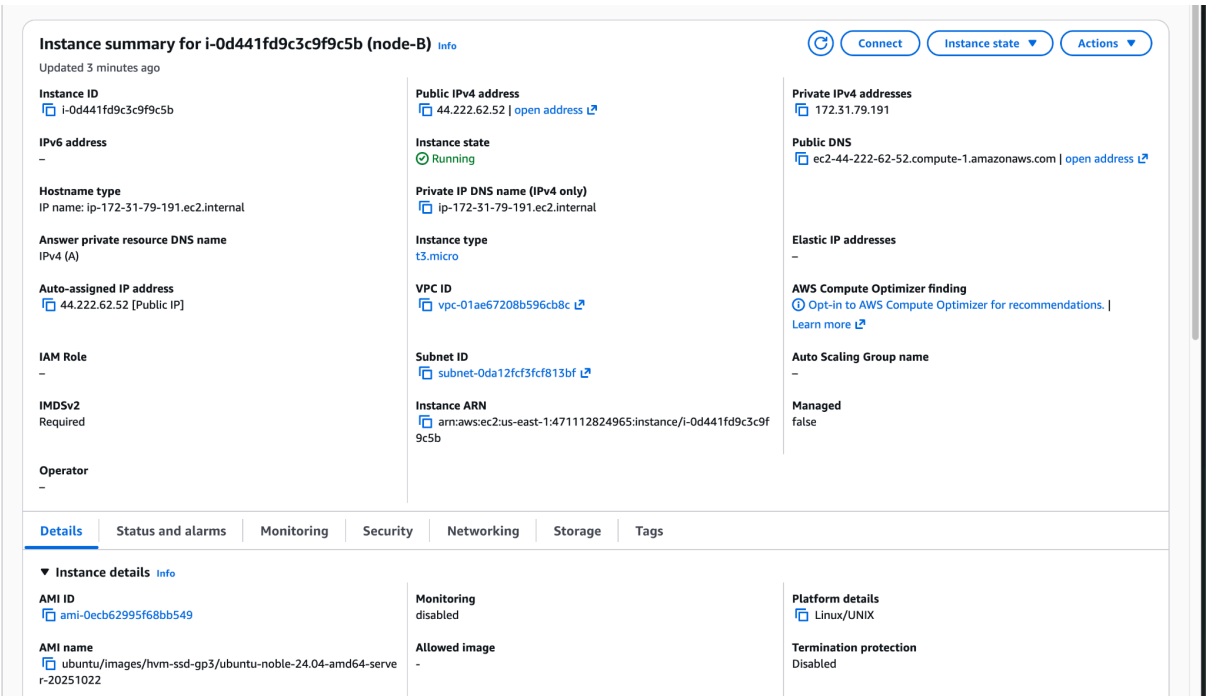
We are running two instances:



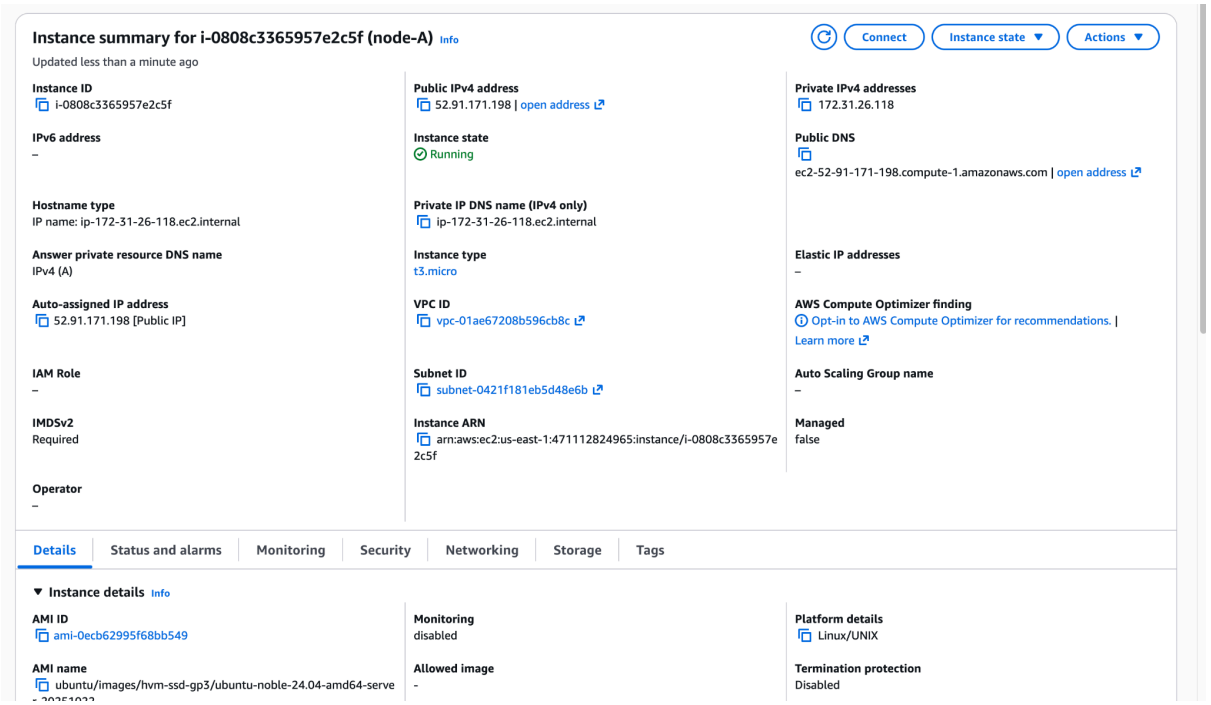

node B is for running the server application, node A is for running the client application

the code for both of the RPC application scripts:
[https://github.com/exekerey/DC3224_assignments/tree/main/lab1](https://github.com/exekerey/DC3224_assignments/tree/main/lab1)

Node B is running on IP 44.222.62.52
We also can see internal IP DNS name that we will use for client - server communication:
ip-172-31-79-191.ec2.internal




Node A is running on IP 52.91.171.198




Connecting via SSH to the servers:

```
danial@Danials-MacBook-Pro ~ % ssh ubuntu@52.91.171.198
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Dec 23 07:43:59 UTC 2025

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   39.7% of 6.71GB   Processes:             108
  Memory usage: 27%               Users logged in:       0
  Swap usage:   0%                IPv4 address for enX0: 172.31.26.118

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

46 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


*** System restart required ***
Last login: Tue Dec 23 07:26:45 2025 from 85.159.27.200
ubuntu@ip-172-31-26-118:~$ ls
client.py  rpc_client.py
```

```
danial@Danials-MacBook-Pro ~ % ssh ubuntu@44.222.62.52
The authenticity of host '44.222.62.52 (44.222.62.52)' can't be established.
ED25519 key fingerprint is SHA256:YX4uMZ0bfmUFVIwLxmfj+cpVJ8BxuV7fU1MRh4K+ljo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.222.62.52' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Dec 23 07:57:26 UTC 2025

  System load:  0.06              Temperature:           -273.1 C
  Usage of /:   26.2% of 6.71GB   Processes:             122
  Memory usage: 23%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.79.191


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Dec 23 07:55:40 2025 from 18.206.107.27
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

Now let’s run the server on the port 5000 that was specified in the lab:

```
ubuntu@ip-172-31-79-191:~$ python3 rpc_server.py
[2025-12-23T08:00:37.277052+00:00] Starting RPC server on 0.0.0.0:5000
```

Running the client application, here are the methods implemented that we could call

```
ubuntu@ip-172-31-26-118:~$ python3 rpc_client.py
Usage:
  python3 rpc_client.py <method> [json_params]
Examples:
  python3 rpc_client.py add '{"a":5,"b":7}'
  python3 rpc_client.py get_time '{}'
  python3 rpc_client.py reverse_string '{"s":"takagi"}'
ubuntu@ip-172-31-26-118:~$
```

and here is the result of calling it:

```
ubuntu@ip-172-31-26-118:~$   python3 rpc_client.py get_time '{}'
{
  "request_id": "f133fbd5-8b60-4d73-a399-4067f96206c1",
  "status": "ok",
  "result": "2025-12-23T08:01:37.208628+00:00",
  "error": null,
  "server_time": "2025-12-23T08:01:37.208632+00:00"
}
ubuntu@ip-172-31-26-118:~$   python3 rpc_client.py add '{"a":5,"b":7}'
{
  "request_id": "317d3e9d-1bfe-453d-823a-e9ed539d4125",
  "status": "ok",
  "result": 12,
  "error": null,
  "server_time": "2025-12-23T08:01:56.465834+00:00"
}
ubuntu@ip-172-31-26-118:~$   python3 rpc_client.py reverse_string '{"s":"takagi"}'
{
  "request_id": "15a8b1b8-fe93-4b4a-b15e-f952e854a222",
  "status": "ok",
  "result": "igakat",
  "error": null,
  "server_time": "2025-12-23T08:02:00.257900+00:00"
}
ubuntu@ip-172-31-26-118:~$
```

We also see processing logs in the server:

```
ubuntu@ip-172-31-79-191:~$ python3 rpc_server.py
[2025-12-23T08:00:37.277052+00:00] Starting RPC server on 0.0.0.0:5000
[2025-12-23T08:01:37.208507+00:00] CONNECT from ('172.31.26.118', 33484)
[2025-12-23T08:01:37.208610+00:00] REQ id=f133fbd5-8b60-4d73-a399-4067f96206c1 method=get_time params={}
[2025-12-23T08:01:37.208660+00:00] RES id=f133fbd5-8b60-4d73-a399-4067f96206c1 status=ok result=2025-12-23T08:01:37.208628+00:00
[2025-12-23T08:01:37.209210+00:00] DISCONNECT from ('172.31.26.118', 33484)
[2025-12-23T08:01:56.465702+00:00] CONNECT from ('172.31.26.118', 42324)
[2025-12-23T08:01:56.465806+00:00] REQ id=317d3e9d-1bfe-453d-823a-e9ed539d4125 method=add params={'a': 5, 'b': 7}
[2025-12-23T08:01:56.465862+00:00] RES id=317d3e9d-1bfe-453d-823a-e9ed539d4125 status=ok result=12
[2025-12-23T08:01:56.466491+00:00] DISCONNECT from ('172.31.26.118', 42324)
[2025-12-23T08:02:00.257770+00:00] CONNECT from ('172.31.26.118', 42332)
[2025-12-23T08:02:00.257870+00:00] REQ id=15a8b1b8-fe93-4b4a-b15e-f952e854a222 method=reverse_string params={'s': 'takagi'}
[2025-12-23T08:02:00.257928+00:00] RES id=15a8b1b8-fe93-4b4a-b15e-f952e854a222 status=ok result=igakat
[2025-12-23T08:02:00.258489+00:00] DISCONNECT from ('172.31.26.118', 42332)
```

Now let’s try making it faulty as required in Task 3.
For this I will block the port 5000 using UFW:

```
ubuntu@ip-172-31-79-191:~$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-79-191:~$ sudo ufw allow 5000/tcp
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-79-191:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
ubuntu@ip-172-31-79-191:~$ exit
logout
Connection to 44.222.62.52 closed.
danial@Danials-MacBook-Pro ~ % ssh ubuntu@44.222.62.52
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Dec 23 08:03:48 UTC 2025

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   26.2% of 6.71GB   Processes:             116
  Memory usage: 23%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.79.191


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Dec 23 07:57:26 2025 from 85.159.27.200
ubuntu@ip-172-31-79-191:~$ sudo ufw deny 5000/tcp
Rule updated
Rule updated (v6)
ubuntu@ip-172-31-79-191:~$
```

Allowing port 22, so SSH connections are still possible
Logging out and connecting again to verify.
And then denying port 5000

And now when we try to call the server, the client timeouts:

```
ubuntu@ip-172-31-26-118:~$   python3 rpc_client.py add '{"a":5,"b":7}'
{
  "request_id": "faaedcb0-efb1-4ac2-be24-dc5cf8419b5a",
  "status": "error",
  "result": null,
  "error": "failed_after_retries: timed out"
}
ubuntu@ip-172-31-26-118:~$
```

Notice how we don't have any logs in the server application, because the request never gets there:

```
ubuntu@ip-172-31-79-191:~$ python3 rpc_server.py
[2025-12-23T08:04:59.173463+00:00] Starting RPC server on 0.0.0.0:5000
```